

Euclid — Book V

Proposition 22

Ex Aequo / First Combination Rule
Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If two chains of magnitudes are proportional term by term, then the first and last terms are proportional ex aequali. In the three-term case:

$$a : b = a' : b', \quad b : c = b' : c' \implies a : c = a' : c'.$$

1 Introduction

Euclid V.22 is the theorem traditionally described as *ex aequali*: when two chains of magnitudes are proportional term by term, the first and last terms are proportional as well.

For the three-term case used throughout the present formal development, the statement is

$$a : b = a' : b', \quad b : c = b' : c' \implies a : c = a' : c'.$$

The current production file is

```
CGJteamLab/Proposition5_22.lean
```

and it is not merely a short wrapper around one theorem. It consolidates three proportion languages:

1. Eudoxus proportions on positive segment classes;
2. raw planar angle-coded segment proportions;
3. raw spatial angle-coded segment proportions.

These layers have different hypotheses and different proof roles.

The Eudoxus theorem is

```
euclid_proposition_5_22
```

and its proof is the compact chain

$$V.16 \longrightarrow V.16 \longrightarrow V.11 \longrightarrow V.16.$$

The raw planar theorem is

```
hilbertSegmentProportionRaw_V22
```

and ultimately depends on the crossing-rays theorem exported by `Proposition4_20.lean`.

The spatial theorem used by XI.27 is

```
hilbertSpaceSegmentProportionRaw_V22_of_crossing
```

and performs the same argument inside one induced plane `PlaneGeo` of the ambient three-space.

Thus the production file is the junction at which Euclid Book V, Hilbert’s Supplement II proportion calculus, the completed Forder circle block, and the later spatial XI.27 construction meet.

2 Euclid V.22: Classical Statement

Euclid’s proposition is stated for any finite chain of magnitudes: if one sequence and another sequence of equal cardinality are in the same ratio taken two by two, then they are also in the same ratio *via equality*, or *ex aequali* [1].

For three terms this becomes

$$a : b = a' : b', \quad b : c = b' : c' \implies a : c = a' : c'.$$

The phrase “via equality” is not an algebraic cancellation rule. In Book V it is a relation between the first and last magnitudes of two chains whose adjacent ratios correspond.

3 Euclid’s Source Proof

Euclid’s source proof is expressed directly in the Eudoxan language of equal multiples.

One chooses arbitrary equal multiples of the first and last magnitudes in the two chains. The pairwise proportion hypotheses transfer the relevant greater-than, equality, and less-than comparisons through the intermediate magnitudes. Proposition V.20 supplies the comparison across the chain. Definition V.5 then converts those transported comparisons back into the proportion between the first and last terms [1].

The source proof therefore has the schematic form

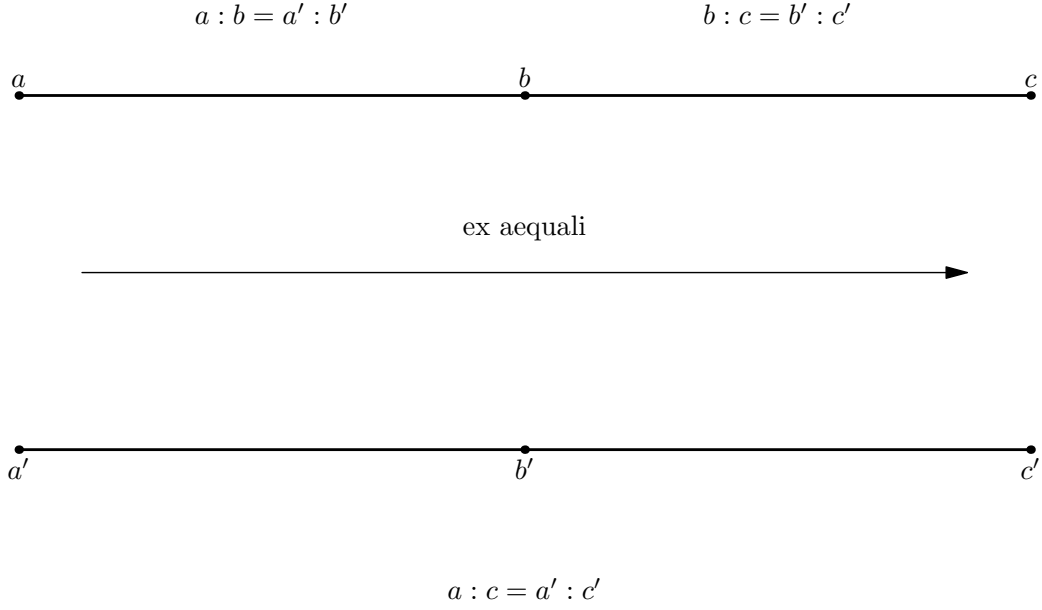


Figure 1: The three-term form of Euclid V.22. The two adjacent proportion equalities share the middle magnitudes b, b' ; ex aequali removes those middle terms and compares the outer pair.

$$\begin{array}{c}
 a : b = a' : b' \\
 b : c = b' : c' \\
 \downarrow \\
 \text{arbitrary equal multiples} \\
 \downarrow \\
 \text{transport of } <, =, > \text{ through the chain} \\
 \downarrow \\
 \text{V.20} \\
 \downarrow \\
 \text{Def. V.5} \\
 \downarrow \\
 a : c = a' : c'.
 \end{array}$$

The current Lean theorem `euclid_proposition_5_22` proves the same three-term statement but uses a different normalization of the classical argument: alternando, transitivity, and alternando again.

The documentation must therefore distinguish the *source proof* from the *production proof*. They prove the same theorem, but they do not have the same internal decomposition.

4 Production Consolidation

The current repository version of `CGJteamLab/Proposition5_22.lean` imports:

```

import CGJteamLab.Proposition5_16
import CGJteamLab.Hilbert3DProportion
import CGJteamLab.HilbertAngleDecomposition
import CGJteamLab.Proposition32
import CGJteamLab.Proposition4_20

```

The file then contains four logical blocks.

1. Raw planar support: right-angle transport, third-angle comparison, raw alternando, and raw V.22 under an explicit crossing-rays hypothesis.
2. Spatial raw support: the same construction performed in a carrier plane of ambient three-space.
3. Closed raw planar theorems: the crossing-rays hypothesis is discharged by the production crossing-rays theorem from Proposition IV.20.
4. Eudoxus V.22: the theorem on positive segment classes, proved from V.16 and V.11.

This consolidation is intentional. The file documents one mathematical principle—composition of proportions—across the proportion languages actually used by the project.

5 Three Proportion Languages

The distinctions between the three layers are fundamental.

5.1 Eudoxus positive-segment proportions

The type

```
HilbertPositiveSegmentClass Geo
```

is the quotient-level magnitude language of Book V.

Proportion is expressed by

```
HilbertEudoxusProportion Geo a b c d
```

and compares arbitrary positive natural multiples through the three possible relations

$$<, \quad =, \quad > .$$

This layer requires the Archimedean plane structure for the substantial Book V theory.

5.2 Raw planar proportions

The raw planar language is geometric rather than quotient-based.

A witness

```
HilbertSegmentRatioWitnessRaw Geo P Q R S
```

contains a proper right triangle AOB with

$$OB \cong PQ, \quad OA \cong RS.$$

The ratio $PQ : RS$ is encoded by the acute angle

$$\angle OAB.$$

Two concrete ratios are proportional when two such witnesses have congruent defining acute angles:

```
def HilbertSegmentProportionRaw ... : Prop :=
  exists w1 : HilbertSegmentRatioWitnessRaw ...,
  exists w2 : HilbertSegmentRatioWitnessRaw ...,
  Geo.AngleCongruent
    w1.0 w1.A w1.B
    w2.0 w2.A w2.B
```

No quotient of segments and no numerical division is used.

5.3 Raw spatial proportions

The spatial language

```
HilbertSpaceSegmentProportionRaw
```

uses the same right-triangle coding, but its segment witnesses live in the ambient three-space.

The crucial restriction is that the proof must not install a global planar `HilbertOrder Geo` or `HilbertCongruence Geo` on the ambient space.

Whenever planar angle geometry is required, the proof chooses an actual carrier plane and works inside

```
PlaneGeo Geo pi
```

before transporting the result back to the ambient geometry.

6 The Eudoxus Theorem

The final Book V theorem is:

```
theorem euclid_proposition_5_22
  [HilbertIncidence Geo]
  [HilbertArchimedeanPlane Geo]
  (a b c a' b' c' : HilbertPositiveSegmentClass Geo)
  (h1 :
    HilbertEudoxusProportion Geo a b a' b')
```

```
(h2 :
  HilbertEudoxusProportion Geo b c b' c') :
  HilbertEudoxusProportion Geo a c a' c'
```

The theorem is exactly the three-term ex-aequali rule.

Its proof contains no circle geometry and no raw ratio witness. It is entirely internal to the already developed Eudoxus theory.

This point is important because the containing module also imports Book IV. That import is needed by the raw proportion blocks, not by the proof body of `euclid_proposition_5_22` itself.

7 First Alternando

From

$$a : b = a' : b'$$

Euclid V.16 gives

$$a : a' = b : b'.$$

In Lean:

```
have h1Alt :
  HilbertEudoxusProportion Geo a a' b b' :=
  euclid_proposition_5_16
  Geo a b a' b' h1
```

The alternation is not syntactic permutation of arguments. It is the already proved proposition V.16 in the Eudoxus magnitude theory.

8 Second Alternando

Apply V.16 to the second hypothesis

$$b : c = b' : c'.$$

This gives

$$b : b' = c : c'.$$

Formally:

```
have h2Alt :
  HilbertEudoxusProportion Geo b b' c c' :=
  euclid_proposition_5_16
  Geo b c b' c' h2
```

The two alternated proportions now share exactly the middle ratio

$$b : b'.$$

9 V.11: Transitivity

Euclid V.11 says that ratios equal to the same ratio are equal to one another.

At the present interface this is simply transitivity of Eudoxus proportion.

From

$$a : a' = b : b'$$

and

$$b : b' = c : c'$$

we obtain

$$a : a' = c : c'.$$

The Lean step is:

```
have hChain :
  HilbertEudoxusProportion Geo a a' c c' :=
  euclid_proposition_5_11
  Geo
  a a'
  b b'
  c c'
  h1Alt
  h2Alt
```

10 Final Alternando

One last use of V.16 transforms

$$a : a' = c : c'$$

into

$$a : c = a' : c'.$$

Thus:

```
exact
  euclid_proposition_5_16
  Geo a a' c c' hChain
```

The entire production proof is therefore

$$(a : b = a' : b') \xRightarrow{V.16} (a : a' = b : b') \xRightarrow{V.11} (a : a' = c : c') \xRightarrow{V.16} (a : c = a' : c')$$

with the second hypothesis alternated in parallel before V.11 is applied.

11 Why the Eudoxus Proof Is So Short

The shortness of `euclid_proposition_5_22` does not mean that the Book V formalization is trivial.

The difficult magnitude work has already been concentrated in:

- the definition of positive segment classes;
- natural multiple arithmetic;
- the Archimedean bridge;
- the Eudoxus comparison clauses;
- V.11;
- V.15;
- V.16.

In particular, V.16 itself is not a free algebraic rearrangement. The current production theorem imports V.11, V.14, and V.15 and proves *alternando* inside the Eudoxus comparison theory.

V.22 is short because the preceding Book V API has reached the correct level of abstraction.

12 Raw Planar Alternando

The raw planar layer cannot use quotient-level Eudoxus transitivity as a substitute for geometry.

Its *alternando* theorem is:

```

theorem hilbertSegmentProportionRaw_alternando_of_crossing_V22
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (hCross :
    HilbertCrossingRaysTransfer (Geo := Geo))
  ...
  (h :
    HilbertSegmentProportionRaw
      Geo
      P Q R S
      U V W X) :
    HilbertSegmentProportionRaw
      Geo
      P Q U V
      R S W X

```

Schematically,

$$PQ : RS = UV : WX \implies PQ : UV = RS : WX.$$

This is Hilbert Supplement II *alternando* for the concrete angle-coded ratio.

13 Cross Placement of the New Legs

Unpack the given raw proportion into two right-triangle witnesses

$$w_1, \quad w_2.$$

The first witness represents $PQ : RS$. Its right triangle has vertex O , denominator ray OA , numerator ray OB , and defining acute angle

$$\angle OAB.$$

The proof now copies the two legs of the second ratio into that same right angle:

$$OC \cong UV$$

with C on ray OA , and

$$OD \cong WX$$

with D on ray OB .

This creates the crossed configuration.

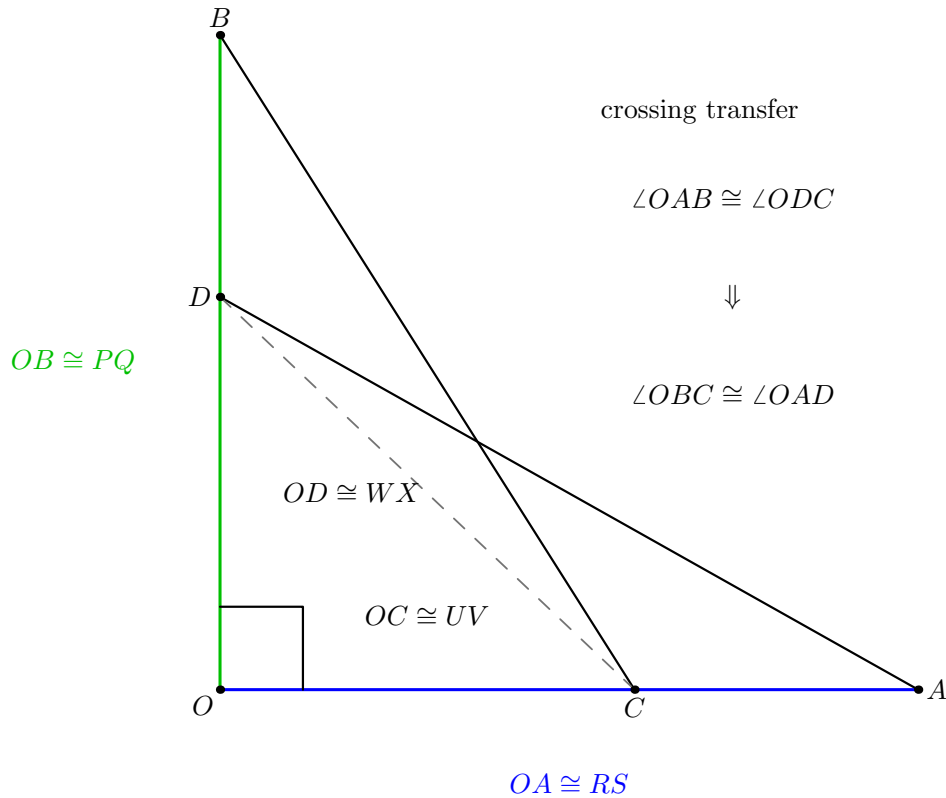


Figure 2: Hilbert's cross-placement for raw alternando. The original right-triangle witness has rays OA, OB . The segment UV is copied to OC on the denominator ray and WX to OD on the numerator ray. The crossing-rays theorem transfers one cross-angle equality to the complementary one.

The copied legs define a new right-triangle witness for $UV : WX$, now inside the geometry of the first witness.

Because all witnesses of the same concrete ratio have congruent defining angles, the original proportion yields

$$\angle OAB \cong \angle ODC.$$

14 The Crossing-Rays Kernel

The crucial input is

```
HilbertCrossingRaysTransfer
```

proved publicly by

```
hilbert_crossing_rays_transfer
```

in `Proposition4_20.lean`.

In the present placement:

$$A, C$$

lie on one ray from O , and

$$D, B$$

lie on the other.

The known cross-angle congruence is

$$\angle OAB \cong \angle ODC.$$

Crossing-rays transfer gives the complementary congruence

$$\angle OBC \cong \angle OAD.$$

This is the exact point where the completed Forder circle theory enters the raw proportion calculus.

15 The Two Alternated Right Triangles

The proof now constructs two new ratio witnesses.

The first right triangle

$$C - O - B$$

represents

$$PQ : UV,$$

because

$$OB \cong PQ, \quad OC \cong UV.$$

The second right triangle

$$D - O - A$$

represents

$$RS : WX,$$

because

$$OA \cong RS, \quad OD \cong WX.$$

The crossing theorem supplied equality of the *other* acute angles of these two right triangles.

The helper

```
hilbert_right_triangle_third_angle_congruent_V22
```

then proves congruence of the remaining acute angles:

$$\angle OCB \cong \angle ODA.$$

Those are precisely the defining angles of the two alternated ratio witnesses.

Hence

$$PQ : UV = RS : WX.$$

16 Right-Triangle Third-Angle Helper

The theorem

```
hilbert_right_triangle_third_angle_congruent_V22
```

says that two proper right triangles with one pair of corresponding acute angles congruent have their remaining acute angles congruent.

The proof is synthetic.

Each triangle is extended and Euclid I.32 produces a decomposition of an adjacent exterior angle. The right angles are congruent; the given acute angles are congruent; angle subtraction then identifies the remaining components.

Thus the raw alternando proof never appeals to

$$\alpha + \beta = 90^\circ.$$

It uses right-angle witnesses, explicit decompositions, and synthetic subtraction.

17 Raw Planar V.22

Once raw alternando has been established, the raw V.22 proof is structurally identical to the compact Eudoxus proof.

The theorem under an explicit crossing hypothesis is

```
hilbertSegmentProportionRaw_V22_of_crossing
```

and begins with

$$a : b = a' : b', \quad b : c = b' : c'.$$

Apply raw alternando:

$$a : a' = b : b',$$

and

$$b : b' = c : c'.$$

Then use raw transitivity:

$$a : a' = c : c'.$$

Apply raw alternando once more:

$$a : c = a' : c'.$$

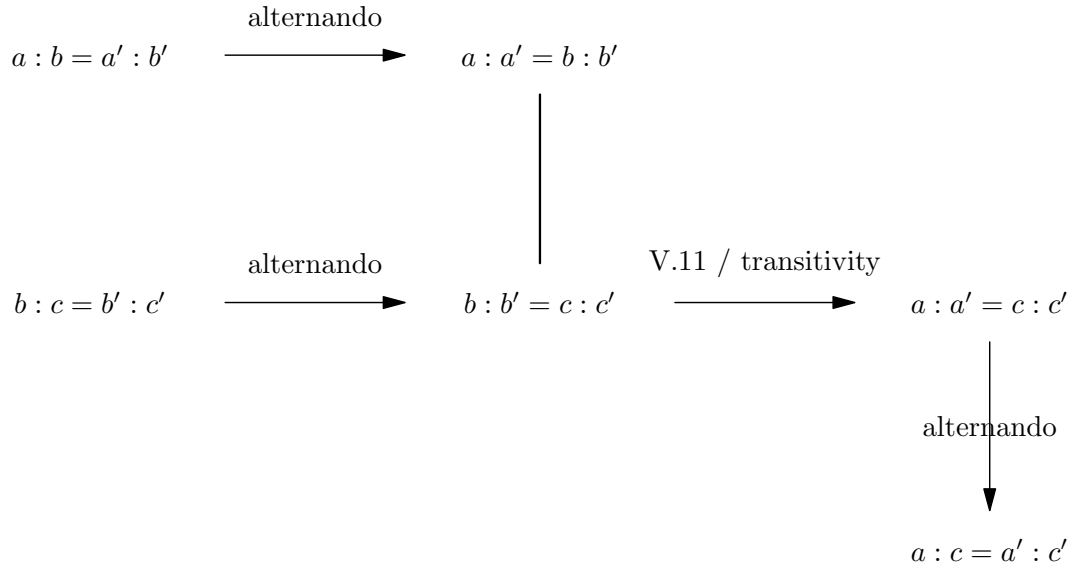


Figure 3: The common proof skeleton of the raw and Eudoxus V.22 theorems: alternando on both hypotheses, transitivity (V.11 at the Eudoxus layer), and one final alternando.

The difference between the layers lies entirely in how alternando and transitivity are realized.

18 Closing the Raw Planar Crossing Hypothesis

The dependency-explicit theorem takes

```
hCross : HilbertCrossingRaysTransfer (Geo := Geo)
```

as an argument.

The production file then exports the closed theorem

```
theorem hilbertSegmentProportionRaw_V22
  [H : HilbertIncidence Geo]
```

```
[HE : @HilbertEuclideanPlane Geo H]
...
```

by supplying

```
hilbert_crossing_rays_transfer Geo
```

from `Proposition4_20.lean`.

The same closure is performed for raw alternando:

```
hilbertSegmentProportionRaw_alternando_V22
```

Thus the final planar raw API has no remaining crossing hypothesis.

19 Why Book IV Appears in V.22

At the quotient-level Eudoxus theorem, V.22 is obtained entirely inside Book V.

At the raw angle-coded level, however, alternando is represented by a geometric transformation of right-triangle ratio witnesses.

That transformation requires the crossing-rays angle theorem.

The production dependency is therefore

Forder IV.12–IV.20 $\longrightarrow$ crossing-rays transfer $\longrightarrow$ raw alternando $\longrightarrow$ raw V.22
--

The presence of `Proposition4_20` in the imports of `Proposition5_22.lean` belongs to this raw layer.

It should not be read as a hidden dependency of the short theorem `euclid_proposition_5_22`.

20 Spatial Crossing-Rays Interface

For ambient three-space the file defines

```
def HilbertSpaceCrossingRaysTransfer ... : Prop :=
  forall pi : S.Plane,
    HilbertCrossingRaysTransfer
      (Geo := PlaneGeo Geo pi)
```

The statement says that every induced plane carries the already proved planar crossing-rays theorem.

This is exactly the form needed by spatial raw alternando.

No global planar structure is imposed on the ambient point type.

21 Spatial Raw Alternando

The theorem is

```
hilbertSpaceSegmentProportionRaw_alternando_of_crossing
```

Its input contains two spatial raw ratio witnesses, which may initially live in unrelated planes.

The proof chooses a carrier plane

$$\pi$$

through the right triangle of the first witness.

The two legs representing the second ratio are copied onto the two rays of that first witness.

Because copied points lie on rays already contained in π , incidence transport proves that the copied points also lie in π .

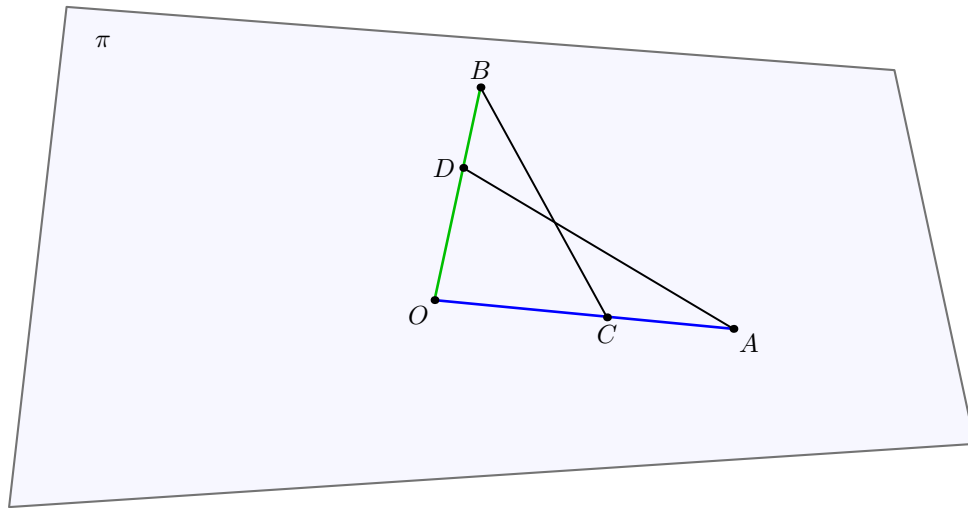
The proof then creates the subtype points

```
Op Ap Bp Cp Dp : PlanePoint Geo pi
```

and transfers all relevant data to

```
PlaneGeo Geo pi.
```

transport result back to ambient space



work inside $\text{PlaneGeo}(\pi)$

Figure 4: Spatial raw alternando chooses the carrier plane π of the first right-triangle witness. The copied legs OC, OD remain in that plane, so the complete crossing-rays argument is performed in PlaneGeo Geo pi and only the result is transported back to ambient three-space.

Inside that induced plane the proof repeats the planar construction:

- right-angle transport;
- copied witness for the second ratio;
- crossing-rays transfer;
- two alternated right triangles;
- third-angle congruence.

Finally, the resulting defining angle congruence is transported back from `PlaneGeo` to the ambient geometry and packaged as `HilbertSpaceSegmentProportionRaw`.

22 Spatial V.22

The spatial first combination rule is:

```
theorem hilbertSpaceSegmentProportionRaw_V22_of_crossing
...
(hCross :
  HilbertSpaceCrossingRaysTransfer
    (Geo := Geo))
...
```

Its proof is again:

alternando + alternando + transitivity + alternando.

The proportion objects are now

```
HilbertSpaceSegmentProportionRaw
```

rather than the planar raw type.

Unlike the planar raw API, the current V.22 module leaves the spatial theorem in dependency-explicit form. It does not export a separate globally closed spatial wrapper.

That closure is performed at the XI.27 integration site.

23 How XI.27 Discharges Spatial Crossing Rays

The production file

```
CGJteamLab/Proposition11_27.lean
```

imports Proposition5_22.lean.

Its theorem

```
hilbertSpaceV22Raw_XI27_of_crossing
```

constructs the plane-wise crossing theorem by

```
have hCross :  
  HilbertSpaceCrossingRaysTransfer  
    (Geo := Geo) := by  
  intro pi  
  exact  
    hilbert_crossing_rays_transfer  
      (PlaneGeo Geo pi)
```

and then calls

```
hilbertSpaceSegmentProportionRaw_V22_of_crossing
```

Thus the spatial closure is explicit and local:

$$\forall \pi, \text{ Book IV crossing theorem in PlaneGeo}(\pi) \implies \text{spatial V.22.}$$

24 Role of V.22 in XI.27

XI.27 constructs a target parallelepiped whose three incident edges must share one coherent scale relative to the source solid.

Two fourth-proportional constructions provide

$$BA : BC = A_0B_0 : A_0K$$

and

$$BC : BF = A_0K : A_0J.$$

V.22 supplies the missing outer relation

$$BA : BF = A_0B_0 : A_0J.$$

This is not a cosmetic consequence.

Without the third ratio, the two independently scaled copied rays have not yet been proved compatible with the fixed target edge A_0B_0 .

V.22 is exactly the synchronization theorem.

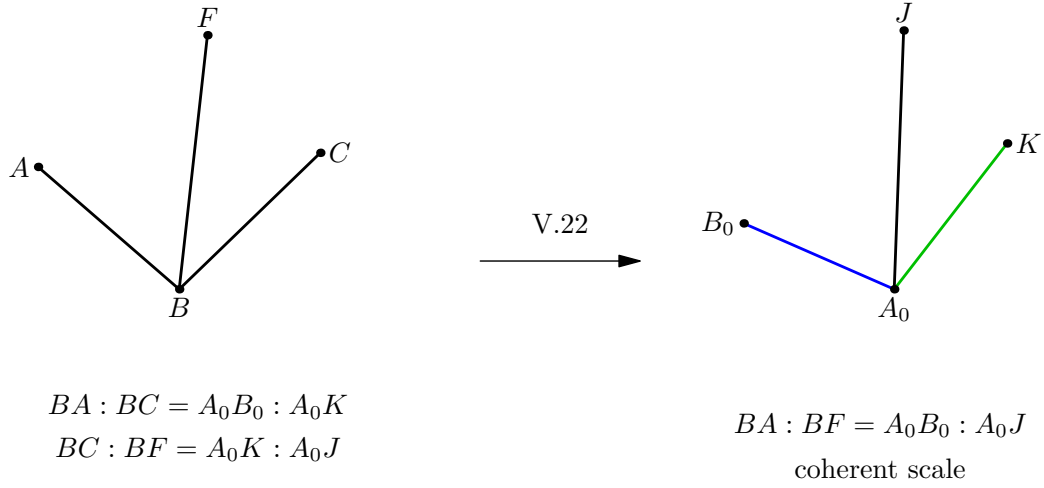


Figure 5: The two fourth-proportional constructions in XI.27 share the middle edge ratio. Spatial V.22 eliminates that middle ratio and supplies the third edge proportion needed to make the copied trihedral corner a single coherent scale.

25 The Two Production Routes to Ex Aequo

The consolidated file therefore contains two conceptually different routes to the same proportion pattern.

25.1 Magnitude route

positive segment classes $\longrightarrow$ Eudoxus proportion $\longrightarrow$ V.16 $\longrightarrow$ V.11 $\longrightarrow$ V.22.

This is the Book V theorem

```
euclid_proposition_5_22
```

25.2 Geometric witness route

right-triangle ratio witnesses $\longrightarrow$ crossing-rays transfer $\longrightarrow$ raw alternando
 $\longrightarrow$ raw transitivity $\longrightarrow$ raw V.22

This is the route used by the spatial XI.27 integration.

The two routes are not definitions of one another in the current production library. They coexist because they serve different parts of the geometry.

26 Relation to V.11 and V.16

At the Eudoxus level, V.22 has only two numbered Book V dependencies.

26.1 V.11

`euclid_proposition_5_11` is transitivity:

$$a : b = c : d, \quad c : d = e : f \implies a : b = e : f.$$

In the present Eudoxus representation its proof is directly `hilbertEudoxusProportion_trans`.

26.2 V.16

`euclid_proposition_5_16` is alternando:

$$a : b = c : d \implies a : c = b : d.$$

Its current production module imports V.11, V.14, and V.15.

Thus the short V.22 theorem inherits the substantial earlier Book V machinery through V.16.

27 Relation to Forder Book IV

The Book IV dependency belongs only to the angle-coded raw route.

The public theorem

```
hilbert_crossing_rays_transfer
```

is exported from

```
CGJteamLab/Proposition4_20.lean
```

after the completion of Forder IV.12–IV.20.

The raw alternando theorem consumes exactly that crossing transfer.

Thus the circle geometry is not an unexplained dependency attached to V.22. It has a sharply localized role: it proves the one cross-angle implication needed to convert one right-triangle ratio placement into its alternated placement.

28 Reusable Helper Layer Inside Proposition5_22

The current production file contains several helpers whose statements are general but whose names retain the V.22 suffix because they were consolidated with this block:

```
hilbert_right_angle_swap_V22  
hilbert_rightAngle_transport_sameRays_V22  
hilbert_right_triangle_third_angle_congruent_V22
```

They support the raw crossing proof.

The first swaps the arms of a proper right angle.

The second transports a right angle when both arms are replaced by points on the same respective rays.

The third proves the remaining acute-angle congruence for two right triangles once one pair of acute angles is known congruent.

These are not new axioms. Their proofs use existing incidence, order, congruence, I.32, and angle-decomposition infrastructure.

29 Axiomatic Profiles

The three V.22 layers do not have the same hypotheses.

29.1 Eudoxus V.22

```
[HilbertIncidence Geo]
[HilbertArchimedeanPlane Geo]
```

The Archimedean structure is inherited from the positive-magnitude Book V development.

29.2 Raw planar V.22

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

The Euclidean plane structure is needed for the right-triangle, angle-decomposition, and crossing-circle machinery.

29.3 Spatial raw V.22

The theorem assumes the spatial incidence/order/congruence hierarchy and the spatial Euclidean structure

```
HilbertSpaceEuclidean Geo
```

Planar Euclidean reasoning is then used only inside induced `PlaneGeo` slices.

This profile is crucial for Book XI, where a global planar order or congruence structure on ambient three-space would be mathematically incorrect.

30 No Temporary Pascal Axiom

The older development explicitly isolated `HilbertCrossingRaysTransfer` so that the proportion calculus could be developed before its circle proof had been completed.

The current production state has closed that gap.

The raw planar production theorems use

```
hilbert_crossing_rays_transfer Geo
```

from the completed Book IV module.

The production comments explicitly record that no temporary axiom from the old Pascal development is imported or used.

This is one of the main foundational milestones represented by the current V.22 file.

31 Production Dependency Graph

The Eudoxus endpoint has the conceptual dependency

V.11
V.14
V.15
↓
V.16
↓
V.22.

More precisely, V.22 imports V.16, while V.16 itself imports V.11, V.14, and V.15.

The raw planar endpoint has

Forder IV.12–IV.20
↓
hilbert_crossing_rays_transfer
↓
raw alternando
↓
raw V.22.

The spatial route continues:

$\forall \pi$, hilbert_crossing_rays_transfer(PlaneGeo π)
↓
spatial raw alternando
↓
spatial raw V.22
↓
XI.27 edge-scale synchronization.

At file level the current production import is:

```
Proposition5_16
Hilbert3DProportion
HilbertAngleDecomposition
Proposition32
Proposition4_20
|
v
Proposition5_22
|
v
Proposition11_27
```

32 Source Proof versus Production Architecture

The phrase “Euclid V.22” therefore names one mathematical principle but several formal realizations.

The source proposition proves ex aequali directly from the Eudoxan comparison of arbitrary multiples.

The current Eudoxus theorem proves the three-term specialization through V.16, V.11, and V.16.

The raw planar theorem realizes alternando geometrically through right-triangle witnesses and crossing rays.

The spatial theorem repeats the raw argument inside a chosen induced plane.

These are not competing proofs accidentally left in one file. They correspond to the distinct proportion languages required by the project.

33 Diagrammatic Audit

Four figures are justified.

1. The abstract ex-aequali chain showing the elimination of the middle pair.
2. The raw crossing construction on the two rays of a right-triangle ratio witness.
3. The three-step production skeleton alternando–transitivity–alternando.
4. The induced-plane spatial construction and the downstream XI.27 synchronization diagram.

The last two ideas are shown in separate figures because the proof-theoretic structure and the spatial application answer different questions.

34 Repository Audit

This document is based on the current repository files on the **master** branch:

```

CGJteamLab/Proposition5_22.lean
CGJteamLab/Proposition5_16.lean
CGJteamLab/Proposition5_11.lean
CGJteamLab/Hilbert3DProportion.lean
CGJteamLab/Proposition4_20.lean
CGJteamLab/Proposition11_27.lean

```

At the documentation checkpoint, the blob SHA of the authoritative V.22 production file is

```
ac4284e833744c6cc9f7e2ef17a40a4f1dbd5074
```

The production source is authoritative if the repository later changes.

35 Conclusion

V.22 is the synchronization theorem of the proportion development.

At the Eudoxus level it is now a very short theorem because V.16 and V.11 already expose the correct abstract algebra of proportion: two alternations align the middle ratio, transitivity removes it, and one final alternation restores the desired orientation.

At the raw geometric level, however, alternando must be constructed. The proof lays two new segment representatives onto the perpendicular rays of a right-triangle ratio witness, invokes the crossing-rays theorem from the completed Forder circle block, and uses a synthetic third-angle theorem to obtain the alternated ratio.

The spatial version performs exactly the same argument inside one induced plane, preserving the distinction between planar geometry and ambient three-space.

This yields the full production chain

Forder IV $\rightarrow$ crossing rays $\rightarrow$ raw alternando $\rightarrow$ spatial V.22 $\rightarrow$ XI.27.
--

In parallel, the quotient-level Book V theorem remains the clean Eudoxus identity

V.16 + V.11 + V.16 $\implies$ V.22.

The current `Proposition5_22.lean` therefore correctly serves as the production junction between the magnitude theory, the synthetic planar ratio calculus, and the three-dimensional application.

References

- [1] Euclid, *Elements, Book V*, translation and commentary by Richard Fitzpatrick.
- [2] David Hilbert, *Grundlagen der Geometrie*, B. G. Teubner, Leipzig, 1899.